

C parity

$$\pi^0 \quad S=0, L=0$$

$$P = (-1)^{L+1} = -1$$

$C_{\pi^0} = ?$ is it eigenstate?

$$\pi^0 \stackrel{?}{=} \bar{\pi}^0$$

$\pi^0 \rightarrow \gamma \gamma$ EM decay conserve C parity

$$\gamma \xleftarrow{C} \pi^0 \xrightarrow{C} \gamma$$

$$\gamma \xleftarrow{C} \pi^0 \xrightarrow{C} \gamma$$

$$C \psi_{\pi^0} = C \psi_{\gamma\gamma} = C_{\gamma} \cdot C_{\gamma} \psi_{\gamma\gamma}$$

$$\Rightarrow C_{\pi^0} = (C_{\gamma})^2 = +1$$

$$\pi^0 \equiv \bar{\pi}^0$$

$$\pi^0 \rightarrow \gamma \gamma$$

$$\pi^0 \rightarrow \gamma \gamma \gamma ?$$

$$C_{\gamma\gamma\gamma} = (C_{\gamma})^3 = (-1)^3 = -1$$

violates C conservation in EM.
experim. not observed

$$\begin{array}{l} B \\ Q \\ L_e \\ L_{\mu} \end{array} \quad \begin{array}{c} \pi^0 \rightarrow \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \quad \begin{array}{cc} e^- & \mu^+ \\ 0 & 0 \\ -1 & +1 \\ 1 & 0 \\ 0 & 1 \end{array}$$

$$\Delta L_e = 1$$

$$\Delta L_{\mu} = 1$$

not observed so far $\Rightarrow$ no lepton flavor violation.

$$\pi^0: L, S, P, C$$

$$J^{PC}$$

$$\vec{J} = \vec{L} \oplus \vec{S}$$

$$L=0, S=0 \Rightarrow J=0$$

$$J^{PC} = 0^{-+}$$

π^0 DECAY MODES

For decay limits to particles which are not established, see the appropriate Search sections (A^0 (axion) and Other Light Boson (X^0) Searches, etc.).

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
$\Gamma_1 \quad 2\gamma$	$(98.823 \pm 0.034) \%$	$S=1.5$
$\Gamma_2 \quad e^+ e^- \gamma$	$(1.174 \pm 0.035) \%$	$S=1.5$
$\Gamma_3 \quad \gamma$ positronium	$(1.82 \pm 0.29) \times 10^{-9}$	
$\Gamma_4 \quad e^+ e^- e^+ e^-$	$(3.34 \pm 0.16) \times 10^{-5}$	
$\Gamma_5 \quad e^+ e^-$	$(6.46 \pm 0.33) \times 10^{-8}$	
$\Gamma_6 \quad 4\gamma$	$< 2 \times 10^{-8}$	CL=90%
$\Gamma_7 \quad \nu \bar{\nu}$	$[a] < 2.7 \times 10^{-7}$	CL=90%
$\Gamma_8 \quad \nu_e \bar{\nu}_e$	$< 1.7 \times 10^{-6}$	CL=90%
$\Gamma_9 \quad \nu_{\mu} \bar{\nu}_{\mu}$	$< 1.6 \times 10^{-6}$	CL=90%
$\Gamma_{10} \quad \nu_{\tau} \bar{\nu}_{\tau}$	$< 2.1 \times 10^{-6}$	CL=90%
$\Gamma_{11} \quad \gamma \nu \bar{\nu}$	$< 6 \times 10^{-4}$	CL=90%

Charge conjugation (C) or Lepton Family number (LF) violating modes

$\Gamma_{12} \quad 3\gamma$	C	$< 3.1 \times 10^{-8}$	CL=90%
$\Gamma_{13} \quad \mu^+ e^-$	LF	$< 3.8 \times 10^{-10}$	CL=90%
$\Gamma_{14} \quad \mu^- e^+$	LF	$< 3.4 \times 10^{-9}$	CL=90%
$\Gamma_{15} \quad \mu^+ e^- + \mu^- e^+$	LF	$< 3.6 \times 10^{-10}$	CL=90%

$$m_{\mu} = 106 \text{ MeV}$$

$$m_e = 0.5 \text{ MeV}$$

$$m_{\pi^0} = 140 \text{ MeV}$$

$$Q > 0$$

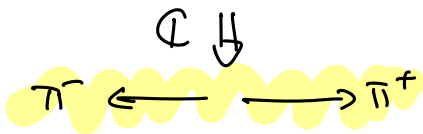
$J=0$: Spinless $S=0$.
ground state $L=0$

$J=1$:
 $S=0$ $L=1$ excited state
 orbital angul. mom.
 $S=1$ $L=0$ ground state of
 a vector particle

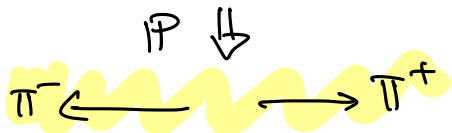
Example 1 Spinless particles

state $\pi^+\pi^- \xrightarrow{C} \pi^-\pi^+$

$$C \psi_{\pi^+\pi^-} = a \psi_{\pi^-\pi^+}$$



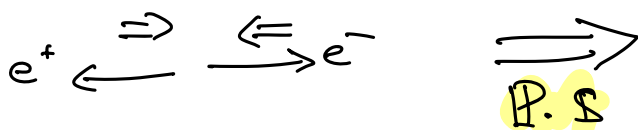
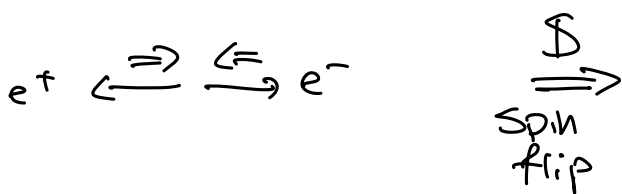
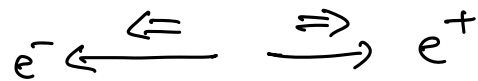
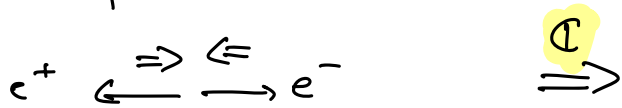
$$P \vec{p} = -\vec{p}$$



$$C \psi_{\pi^+\pi^-} \equiv P \psi_{\pi^+\pi^-} = (-1)^L \psi_{\pi^+\pi^-}$$

L : total ang. mom
of $\pi^+\pi^-$ state

Example 2 $S=1/2$ particles e^+e^-



$$C \equiv P.S$$

same effect on e^+e^-

$$P \psi_{e^+e^-} = (-1)(-1)^L \psi_{e^+e^-}$$

↪ intrinsic parity

$$S=0 = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

$$S \psi_{e^+e^-} = (-1)^{S+1} \psi_{e^+e^-}$$

$$S=1 = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

S : total spin of e^+e^-

$$\begin{aligned} C \psi_{e^+e^-} &= (P \cdot S) \psi_{e^+e^-} = (-1)(-1)^L (-1)^{S+1} \psi_{e^+e^-} \\ &= (-1)^{L+S} (-1)^2 \psi_{e^+e^-} = (-1)^{L+S} \psi_{e^+e^-} \end{aligned}$$

L : total ang. mom of state e^+e^-

S : total spin of the state e^+e^-

general for a state with $P\bar{P}$

$$\psi_{P\bar{P}} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{intrinsic}}$$

Fermions: $P\bar{P} \equiv f\bar{f}$

$$C \psi_{\text{space}} \equiv P \psi_{\text{space}} = (-1)^L \psi_{\text{space}}$$

$$C \psi_{\text{spin}} \equiv S \psi_{\text{spin}} = (-1)^{S+1} \psi_{\text{spin}}$$

$$C \psi_{\text{intrinsic}} = -1 \quad \text{from QFT}$$

$$\Rightarrow C_{f\bar{f}} = (-1)^2 (-1)^{L+S} = (-1)^{L+S}$$

Bosons: $P\bar{P} \equiv b\bar{b}$

$$C \psi_{\text{space}} \equiv P \psi_{\text{space}} = (-1)^L \psi_{\text{space}}$$

$$C \psi_{\text{spin}} \equiv S \psi_{\text{spin}} = (-1)^S \psi_{\text{spin}}$$

$$C \psi_{\text{intrinsic}} = +1 \quad \text{for } b\bar{b} \text{ from QFT}$$

$$C \psi_{b\bar{b}} = (-1)^L (-1)^S (+1) = (-1)^{L+S}$$

$$\Rightarrow \text{for any } P\bar{P} \text{ state } C = (-1)^{L+S}$$

Exercise:

$$P\bar{P} \pi^+ \pi^-$$

$$P P P \bar{P}$$

$$\pi^+ \pi^- \pi^0$$

$$P P \pi^+ \pi^-$$

$$\pi^0 \pi^0 \pi^0$$

$$\pi^+ \pi^- \pi^+ \pi^-$$

G - Parity

very few eigenstates for $G \Rightarrow$ not very practical.

In strong interactions: C and I are conserved.
isospin.

$\Rightarrow$ new operator $G = C \times R_2$

C : Charge parity

$$R_2 = e^{i\pi I_2}$$

Rotation around $\hat{I}_2$ in isospin space

$$C|\pi^+\rangle = \alpha|\pi^-\rangle$$

$\pi^\pm, \pi^0$: $I=1$ triplet

$$|\pi^-\rangle = |1, -1\rangle$$

$$|\pi^0\rangle = |1, 0\rangle$$

$$|\pi^+\rangle = |1, +1\rangle$$

$$R_2|\pi^+\rangle = (-1)^{I-I_3}|\pi^-\rangle$$

$$= (-1)^{1-1}|\pi^-\rangle$$

$$= |\pi^-\rangle$$

$$G|\pi^0\rangle = C R_2|\pi^0\rangle = C (-1)^{1-0}|\pi^0\rangle = C\pi^0 (-1)^1|\pi^0\rangle$$

$$= -|\pi^0\rangle$$

$G_\pi = -1$ for π triplet

$$G|\pi^+\rangle = -|\pi^+\rangle$$

G parity for $I=1$ triplet

$$G|\pi^-\rangle = -|\pi^-\rangle$$

$$R_2|\pi^0\rangle = (-1)^I|\pi^0\rangle$$

$$R_2 = (-1)^I$$

for triplet elements

$$\Rightarrow G = (-1)^{L+S} (-1)^I = (-1)^{L+S+I}$$

eigenvalue for neutral element of triplet

G parity conserved for strong interactions.

G defined only isospin multiplets w/ total electric charge = 0

I $\rightarrow$ related to electric charge

Gell-Mann - Nishijima formula for electric charge
nucleus ${}^A_Z N$

$$I_3 = Z \times \frac{1}{2} + (A-Z) \times \left(-\frac{1}{2}\right) = Z \frac{1}{2} + Z \frac{1}{2} - \frac{A}{2}$$

$$= Z - \frac{A}{2}$$

$$\Rightarrow Z = I_3 + \frac{A}{2}$$

$\downarrow$

$$Q = I_3 + \frac{B}{2}$$

B: baryon number

$$\pi^+ : B=0 \quad Q = I_3 + \frac{0}{2} = +1 \quad |\pi^+\rangle = |1, 1\rangle$$

Strange particles: $K^\pm, K^0$

$$Q = I_3 + \frac{B + \text{Strangeness}}{2}$$

Name	$\pi^\pm$	π^0	$K^\pm$	K^0	η	p	n	Λ	$\Sigma^{\pm,0}$	Δ
Mass (MeV)	140	135	494	498	548	938	940	1116	1190	1232
Charge	± 1	0	± 1	0	0	1	0	0	$\pm 1, 0$	$2, \pm 1, 0$
Parity	-	-	-	-	-	+	+	+	+	+
Baryon n.	0	0	0	0	0	1	1	1	1	1
Spin	0	0	0	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{2}$

Eight fold way $\rightarrow$ Static quark model

Developed by
independently

Gell-Mann
Zweig

1961-1968

Isospin, Spin, Strangeness

$SU(N)$ group of transformations

unitary matrices of $N \times N$

$$\det = 1$$

$$\text{trace} = 0$$

$$U \in SU(N) \quad U = e^{i a_j T_j}$$

$$j = 1, \dots, N^2 - 1$$

$N^2 - 1$: generators

T_j : generators.

a_j : parameters

$N - 1$: diagonal generators

$SU(2)$: 3 generators S_x, S_y, S_z I_1, I_2, I_3
1 diagonal gen. S_z I_3

Static quark model : hadrons are multiplets of $SU(3)$

$SU(3)$: 8 generators

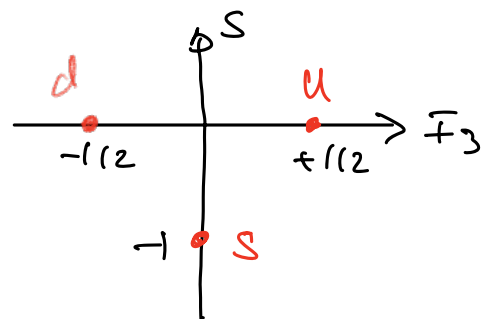
2 diagonal : I_3 , strangeness

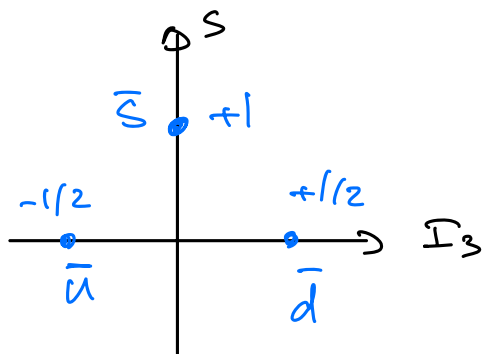
Fundamental repp. of $SU(3)$ made of quarks

$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad d = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad S = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

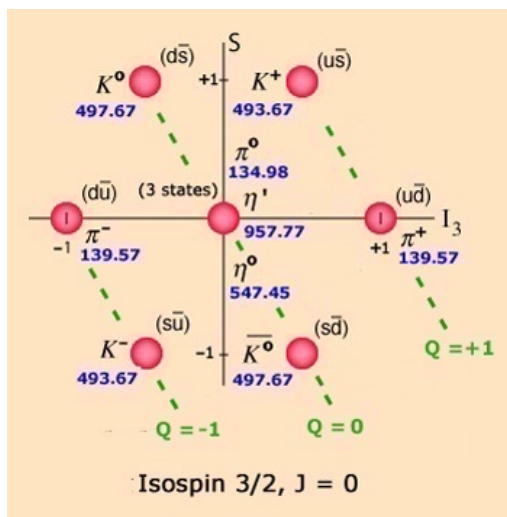
$$\textcircled{3} \quad u, d, s \quad \textcircled{\bar{3}} \quad \bar{u}, \bar{d}, \bar{s}$$

	I_3	S
u	$+1/2$	0
d	$-1/2$	0
s	0	-1





proton = uud
 $B = 1 \Rightarrow B = 1/3$ for quarks.
 $B = -1/3$ for $\bar{u}, \bar{d}, \bar{s}$

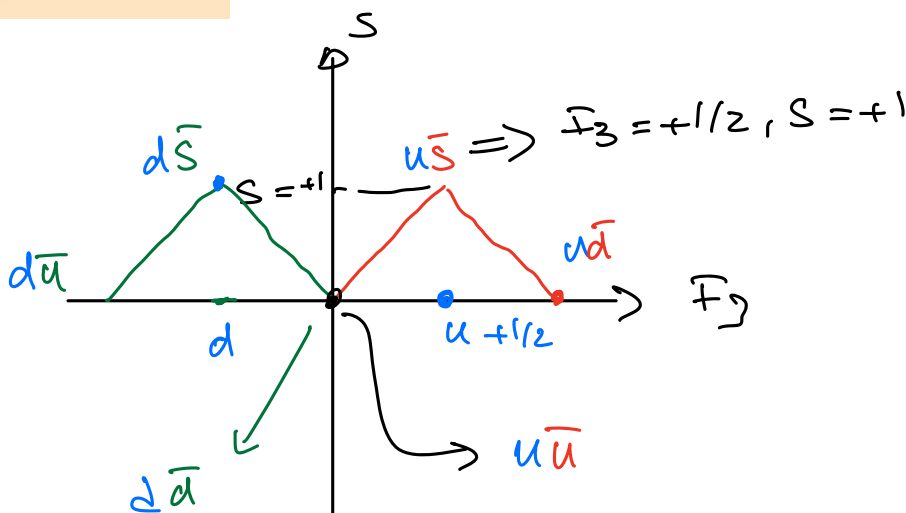
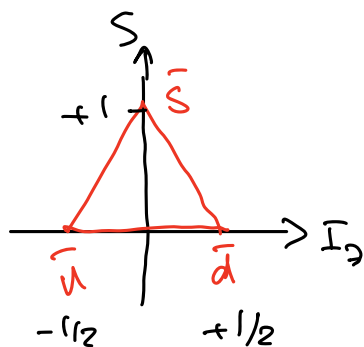


Mesons made of $q, \bar{q}$

$$\text{Mesons} = 3 \otimes \bar{3} = 1 \oplus 8$$

$u \times (\bar{u}, \bar{d}, \bar{s})$

$d \times (\bar{u}, \bar{d}, \bar{s})$



$u\bar{s} :$
 $||$
 K^+

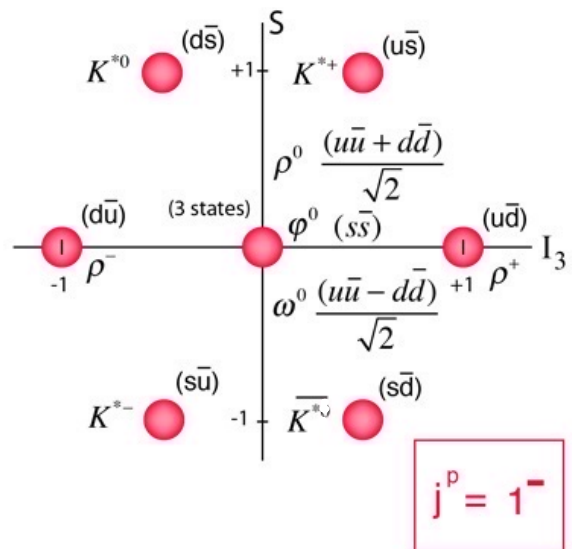
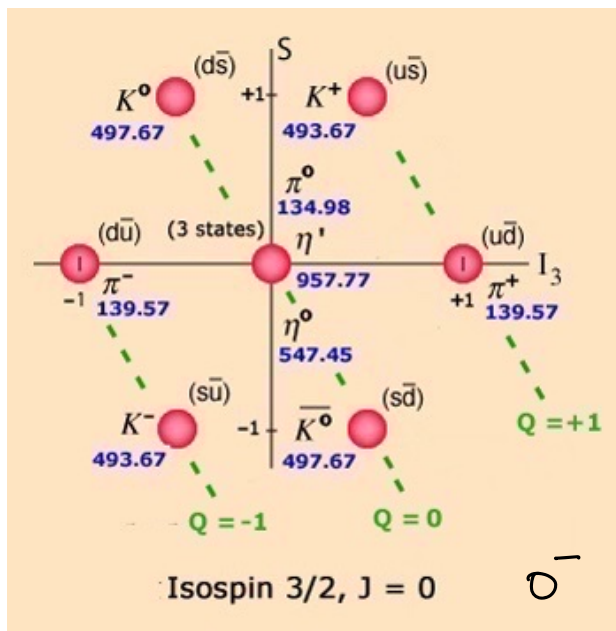
$$Q = I_3 + \frac{B+S}{2} = +1/2 + \frac{1}{2} \left(\frac{1}{3} - \frac{1}{3} + 1 \right) = +1$$

Quarks: $S = 1/2$ particles.

Mesons: $S = 0, S = 1$

$S = 0$ scalar mesons

$S = 1$ vector mesons.

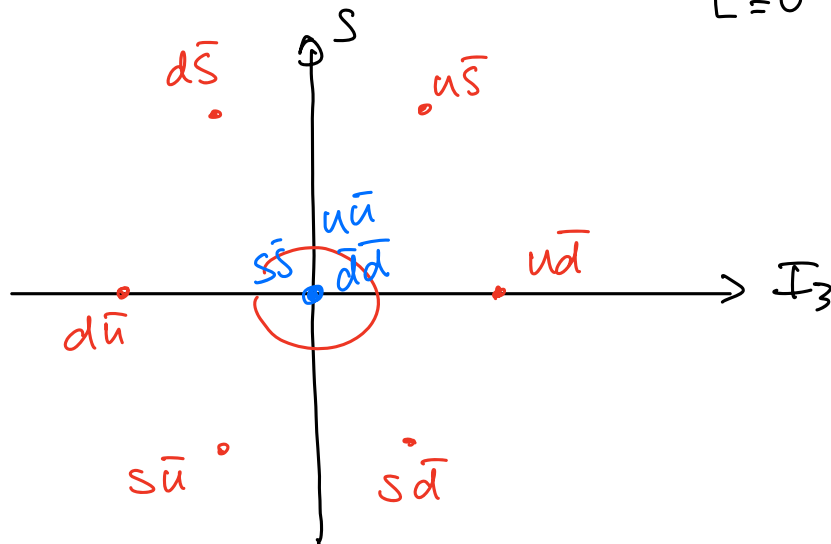


$$S=0$$

$$L=0$$

$$S=1$$

$$L=0 \quad \left. \vphantom{\begin{matrix} S=1 \\ L=0 \end{matrix}} \right\} J=1$$



$$u\bar{u}, s\bar{s}, d\bar{d} : \quad I_3 = 0 \rightarrow I = 0, 1$$

$$S_3 = 0 \rightarrow S = 0, 1$$

$$\text{Singlet} \quad \psi_0 = \frac{1}{\sqrt{3}} (u\bar{u} + d\bar{d} + s\bar{s})$$

$$\left. \begin{array}{l} \text{Flavor} \\ \text{octet} \end{array} \right\} \begin{array}{l} \psi_{8,1} = \frac{1}{\sqrt{2}} (u\bar{u} - d\bar{d}) \quad \pi^0 \\ \psi_{8,0} = \frac{1}{\sqrt{6}} (u\bar{u} + d\bar{d} - 2s\bar{s}) \end{array} \quad \begin{array}{l} I_3 = 0 \quad I = 1 \\ I_3 = 0 \quad I = 0 \end{array}$$

$$\left. \begin{array}{l} \text{Flavor} \\ \text{singlet} \end{array} \right\} \psi_{1,0} = \frac{1}{\sqrt{3}} (u\bar{u} + d\bar{d} + s\bar{s}) \quad I_3 = 0 \quad I = 0.$$

$a_i \bar{a}_j$

$q_i :: \text{spin } 1/2$

$L=0$ ground state

$L=0$

$S=0$

pseudo-scalar mesons.

$\uparrow\downarrow$

$S=1$

vector mesons.

$\uparrow\uparrow \quad \downarrow\downarrow$

$$\left. \begin{aligned} \pi^0(140) &\approx \psi_{8,1}^{ps} = (u\bar{u} - d\bar{d})/\sqrt{2} \\ \eta(550) &= \psi_{8,0}^{ps} \cos\theta_{ps} - \psi_1^{ps} \sin\theta_{ps} \\ \eta'(960) &= \psi_{8,0}^{ps} \sin\theta_{ps} + \psi_1^{ps} \cos\theta_{ps} \end{aligned} \right\} \begin{aligned} J^P &= 0^-, \\ \theta_{ps} &\approx -25^\circ. \end{aligned}$$

$$\left. \begin{aligned} \rho^0(770) &\approx \psi_{8,1}^v = (u\bar{u} - d\bar{d})/\sqrt{2} \\ \phi(1020) &= \psi_{8,0}^v \cos\theta_v - \psi_1^v \sin\theta_v \approx s\bar{s} \\ \omega(780) &= \psi_{8,0}^v \sin\theta_v + \psi_1^v \cos\theta_v \approx (u\bar{u} + d\bar{d})/\sqrt{2} \end{aligned} \right\} \begin{aligned} J^P &= 1^-, \\ \theta_{vect} &\approx 36^\circ \end{aligned}$$

